

Patient 12 Tumor: Raw Pixel Intensity Distribution

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2026-06-01

Load libraries

```
library(ggplot2)
library(ggribes)
library(dplyr)

##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(reshape2)
```

See what files exist

```
list.files('/restricted/projectnb/camplab/projects/20220504_Suzuki_LN/Data/Imaging/IMC/Tumor/PardeeTumor')

## [1] "Patient_12T_ROI_001_1.txt" "Patient_12T_ROI_002_2.txt"
## [3] "Patient_12T_ROI_003_3.txt" "Patient_12T_ROI_004_4.txt"
## [5] "Patient_12T.mcd"
```

Load raw pixel data

```
data_dir <- "/restricted/projectnb/camplab/projects/20220504_Suzuki_LN/Data/Imaging/IMC/Tumor/PardeeTumor"

txt_files <- list.files(data_dir, pattern = "\\\\.txt$", full.names = TRUE)
cat("Found", length(txt_files), "ROI files:\n")

## Found 4 ROI files:
cat(basename(txt_files), sep = "\n")

## Patient_12T_ROI_001_1.txt
## Patient_12T_ROI_002_2.txt
## Patient_12T_ROI_003_3.txt
## Patient_12T_ROI_004_4.txt
```

```
# Read one ROI first to inspect structure
roi1 <- read.delim(txt_files[1], check.names = FALSE)
cat("\nDimensions:", nrow(roi1), "pixels x", ncol(roi1), "columns\n")

##
## Dimensions: 1000000 pixels x 47 columns
cat("Column names:\n")

## Column names:
print(colnames(roi1))

## [1] "Start_push"      "End_push"        "Pushes_duration"
## [4] "X"              "Y"              "Z"
## [7] "SMA(Pr141Di)"    "CD19(Nd142Di)"   "TIM3(Nd143Di)"
## [10] "EpCAM(Nd144Di)"  "TBet(Nd145Di)"   "RORgT(Nd146Di)"
## [13] "CD163(Sm147Di)"  "CCR7(Nd148Di)"   "CD15(Sm149Di)"
## [16] "PDL1(Nd150Di)"   "GATA3(Eu151Di)"  "CD56(Sm152Di)"
## [19] "LAG3(Eu153Di)"   "CD1c(Sm154Di)"   "CD86(Gd155Di)"
## [22] "CD4(Gd156Di)"    "CD103(Gd158Di)"  "CD68(Tb159Di)"
## [25] "CD8a(Gd160Di)"   "CD11c(Dy161Di)"  "FoxP3(Dy162Di)"
## [28] "LAMP3(Dy163Di)"  "CD117(Dy164Di)"  "PD1(Ho165Di)"
## [31] "CD45RA(Er166Di)" "CD39(Er167Di)"   "CD138(Er168Di)"
## [34] "TIGIT(Tm169Di)"  "CD141(Er170Di)"  "HLADRB(Yb171Di)"
## [37] "MKI67(Yb172Di)"  "CD45RO(Yb173Di)" "CD28(Yb174Di)"
## [40] "CD14(Lu175Di)"   "CTLA4(Yb176Di)"  "DNA191(Ir191Di)"
## [43] "DNA193(Ir193Di)" "TTF1orCD16(Pt194Di)" "CD127(Pt195Di)"
## [46] "Membrane(Pt196Di)" "CD38(Pt198Di)"
```

Take a quick look at the data

```
head(roi1)

##   Start_push End_push Pushes_duration X Y Z SMA(Pr141Di) CD19(Nd142Di)
## 1      1866    2249              384 0 0 0      22.214      1.020
## 2      2252    2635              384 1 0 1      10.472      2.390
## 3      2637    3020              384 2 0 2      15.284      4.000
## 4      3022    3405              384 3 0 3       8.421      1.916
## 5      3407    3790              384 4 0 4      35.065      2.000
## 6      3793    4176              384 5 0 5      20.091      2.000
##   TIM3(Nd143Di) EpCAM(Nd144Di) TBet(Nd145Di) RORgT(Nd146Di) CD163(Sm147Di)
## 1      22.678      1.000      3.027      0.000      0.000
## 2      20.306      0.000      1.370      1.248      0.000
## 3      28.531      2.428      4.245      1.000      2.878
## 4      17.129      1.149      0.000      0.000      0.000
## 5      24.545      0.000      2.646      1.498      0.000
## 6      21.076      0.000      0.000      0.000      0.000
##   CCR7(Nd148Di) CD15(Sm149Di) PDL1(Nd150Di) GATA3(Eu151Di) CD56(Sm152Di)
## 1      0.000      8.503      4.949      0.00      2.000
## 2      1.035      4.339      1.000      1.58      2.059
## 3      4.854      3.522      0.000      0.00      0.000
## 4      0.000      3.358      0.000      0.00      0.000
## 5      1.255      3.000      2.000      0.00      1.330
## 6      0.000      2.038      1.000      0.00      0.000
```

##	LAG3(Eu153Di)	CD1c(Sm154Di)	CD86(Gd155Di)	CD4(Gd156Di)	CD103(Gd158Di)
## 1	0	62.983	1.00	5.511	5.597
## 2	0	76.561	1.00	2.000	11.083
## 3	1	44.971	0.00	10.287	4.250
## 4	0	27.357	0.00	5.071	8.790
## 5	0	27.284	0.00	6.750	2.282
## 6	0	23.612	2.17	11.893	7.407
##	CD68(Tb159Di)	CD8a(Gd160Di)	CD11c(Dy161Di)	FoxP3(Dy162Di)	LAMP3(Dy163Di)
## 1	12.929	0.000	1.101	3.538	2.343
## 2	25.730	3.324	1.166	5.526	1.000
## 3	21.022	2.421	2.781	9.849	1.000
## 4	12.054	2.000	4.335	6.037	0.000
## 5	13.442	0.000	1.000	7.706	2.000
## 6	7.864	1.000	3.497	3.791	1.000
##	CD117(Dy164Di)	PD1(Ho165Di)	CD45RA(Er166Di)	CD39(Er167Di)	CD138(Er168Di)
## 1	0.881	4.21	4.228	0.000	5.215
## 2	0.000	1.00	24.073	2.183	7.338
## 3	0.000	1.00	24.843	2.741	10.223
## 4	0.000	0.00	13.749	0.000	5.424
## 5	0.718	0.00	21.237	0.000	29.016
## 6	0.000	0.00	25.729	0.000	12.989
##	TIGIT(Tm169Di)	CD141(Er170Di)	HLADRB(Yb171Di)	MKI67(Yb172Di)	CD45RO(Yb173Di)
## 1	1.000	7.782	68.487	1.230	133.350
## 2	0.000	1.000	107.447	5.363	123.187
## 3	0.000	2.125	138.068	12.249	74.044
## 4	0.000	5.293	89.632	3.662	19.094
## 5	0.000	6.260	156.163	11.446	21.224
## 6	1.056	1.000	128.824	3.000	19.993
##	CD28(Yb174Di)	CD14(Lu175Di)	CTLA4(Yb176Di)	DNA191(Ir191Di)	DNA193(Ir193Di)
## 1	9.677	14.753	0.000	221.781	412.416
## 2	16.223	27.528	0.000	272.491	514.695
## 3	14.421	34.252	0.000	283.317	490.711
## 4	3.202	19.113	1.141	212.300	390.267
## 5	11.316	29.465	0.000	319.015	544.005
## 6	14.750	12.222	0.000	145.415	218.283
##	TTF1orCD16(Pt194Di)	CD127(Pt195Di)	Membrane(Pt196Di)	CD38(Pt198Di)	
## 1	9.852	6.431	57.334	4.234	
## 2	4.859	4.226	39.337	8.444	
## 3	15.032	6.000	71.685	6.645	
## 4	14.278	8.854	46.024	4.434	
## 5	11.656	8.188	78.105	7.208	
## 6	4.237	11.934	84.994	6.187	

The first 6 columns are metadata and coordinates. Columns 7-47 are the 41 marker channels. Each row is one pixel.

Load ROIs and sample a subset

```
# Columns 1-6 are metadata, 7 onwards are markers
marker_cols <- colnames(roi1)[7:ncol(roi1)]

# Load all ROIs, sample 50k pixels each to keep it manageable
set.seed(42)
```

```

all_rois <- lapply(seq_along(txt_files), function(i) {
  df <- read.delim(txt_files[i], check.names = FALSE)
  df_markers <- df[, marker_cols]
  df_sample <- df_markers[sample(nrow(df_markers), 50000), ]
  df_sample$ROI <- paste0("ROI_", i)
  df_sample
})

pixels <- do.call(rbind, all_rois)
cat("Total pixels loaded:", nrow(pixels), "\n")

```

```
## Total pixels loaded: 200000
```

Reshape to long format and apply asinh transformation

```

# Reshape to long format
pixels_long <- melt(pixels, id.vars = "ROI",
                    variable.name = "marker",
                    value.name = "intensity")

# Apply asinh transformation once so plots and summary table are consistent
pixels_long <- pixels_long %>%
  mutate(intensity_asinh = asinh(intensity / 1))

```

Histograms: Pixel Intensity Landscape (ROI 1)

```

roi1_data <- pixels_long %>%
  filter(ROI == "ROI_1")

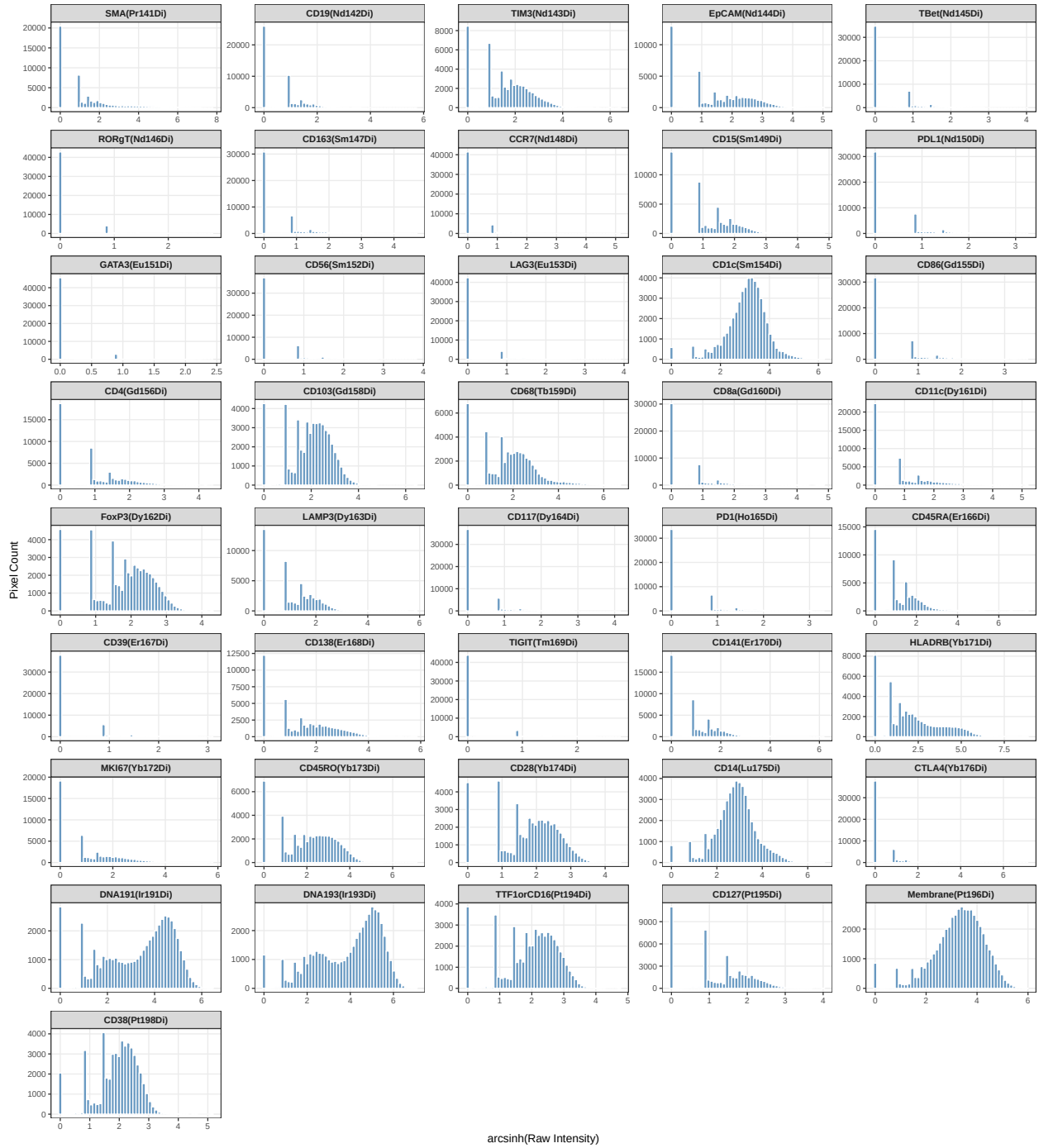
ggplot(roi1_data, aes(x = intensity_asinh)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white", size = 0.1, alpha = 0.8) +
  facet_wrap(~ marker, scales = "free", ncol = 5) +
  labs(
    title = "Pixel Intensity Landscape: Patient 12 (ROI 1)",
    subtitle = "Histogram distribution of all 41 markers",
    x = "arcsinh(Raw Intensity)",
    y = "Pixel Count"
  ) +
  theme_bw(base_size = 11) +
  theme(
    strip.text = element_text(face = "bold", size = 9),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(size = 8),
    axis.text.y = element_text(size = 8)
  )

```

```
## Warning in geom_histogram(bins = 50, fill = "steelblue", color = "white", :
## Ignoring unknown parameters: `size`
```

Pixel Intensity Landscape: Patient 12 (ROI 1)

Histogram distribution of all 41 markers



Density Ridges: Per-ROI Pixel Intensity Distributions (All Markers)

Each ridge represents one ROI. This makes it easy to compare distributions across ROIs and spot any ROI-level artifacts.

```
ggplot(pixels_long, aes(x = intensity_asinh, y = ROI, fill = ROI)) +  
  # scale < 1.0 guarantees ridges never physically overlap  
  geom_density_ridges(alpha = 0.8, scale = 0.9, rel_min_height = 0.01, show.legend = FALSE) +
```

```

facet_wrap(~ marker, scales = "free_x", ncol = 5) +
labs(
  title      = "Patient 12 - Pixel Intensity Distributions per Marker",
  subtitle   = "Arcsinh-Transformed, Cofactor = 1",
  x          = "arcsinh(Raw Intensity)",
  y          = "Region of Interest (ROI)"
) +
theme_ridges(font_size = 10) +
theme(
  strip.text      = element_text(face = "bold", size = 9),
  axis.text.y     = element_text(size = 8),
  axis.title      = element_text(size = 11, face = "bold"),
  axis.title.x    = element_text(hjust = 0.5),
  axis.title.y    = element_text(hjust = 0.5, angle = 90)
)

```

```

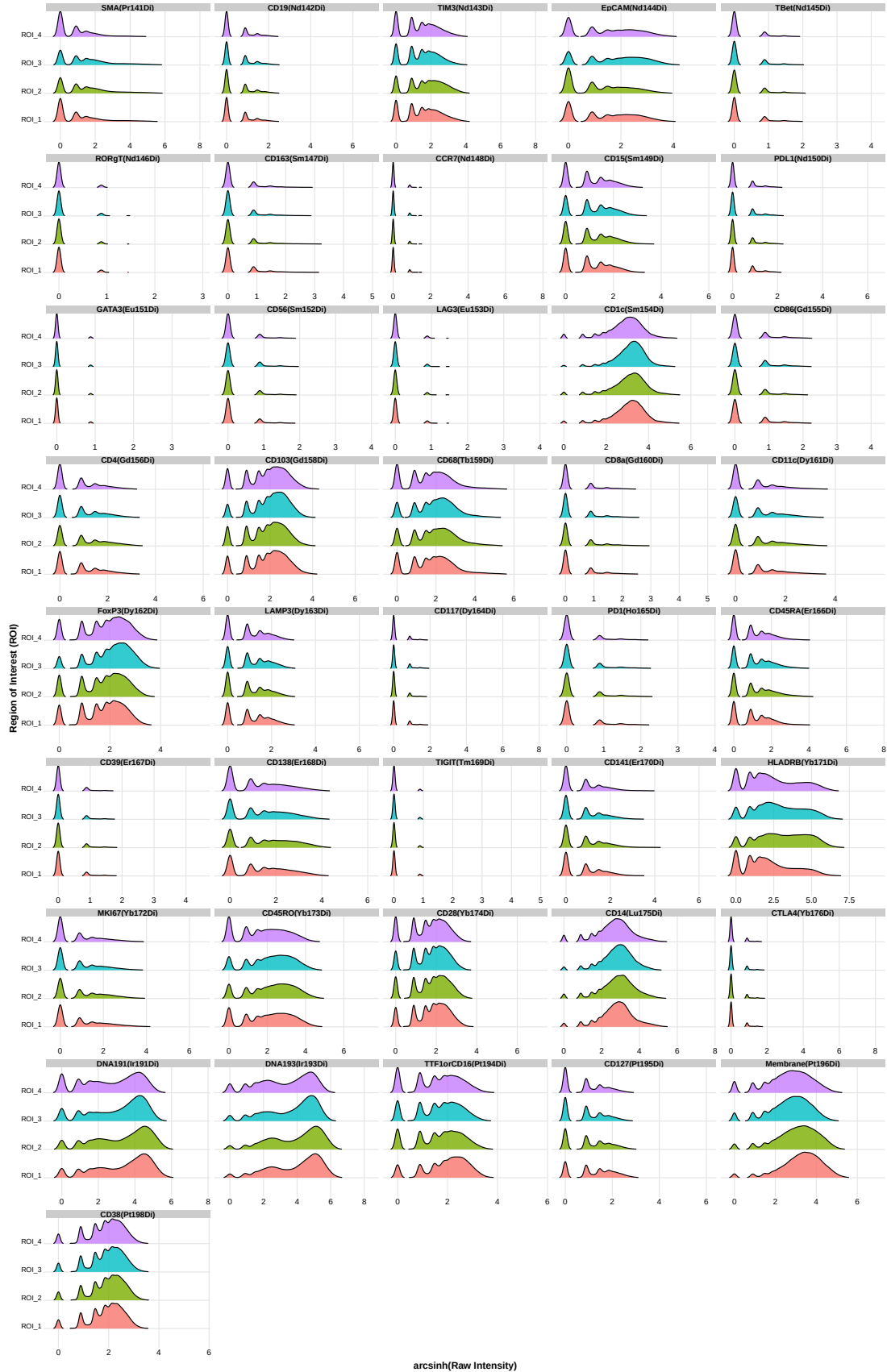
## Picking joint bandwidth of 0.136
## Picking joint bandwidth of 0.0737
## Picking joint bandwidth of 0.103
## Picking joint bandwidth of 0.115
## Picking joint bandwidth of 0.0598
## Picking joint bandwidth of 0.04
## Picking joint bandwidth of 0.0692
## Picking joint bandwidth of 0.0447
## Picking joint bandwidth of 0.0874
## Picking joint bandwidth of 0.0657
## Picking joint bandwidth of 0.0312
## Picking joint bandwidth of 0.0566
## Picking joint bandwidth of 0.0423
## Picking joint bandwidth of 0.0697
## Picking joint bandwidth of 0.066
## Picking joint bandwidth of 0.0934
## Picking joint bandwidth of 0.0876
## Picking joint bandwidth of 0.109
## Picking joint bandwidth of 0.0713
## Picking joint bandwidth of 0.0992
## Picking joint bandwidth of 0.0801
## Picking joint bandwidth of 0.0872
## Picking joint bandwidth of 0.0553
## Picking joint bandwidth of 0.0659
## Picking joint bandwidth of 0.101

```

Picking joint bandwidth of 0.053
Picking joint bandwidth of 0.118
Picking joint bandwidth of 0.0377
Picking joint bandwidth of 0.0987
Picking joint bandwidth of 0.172
Picking joint bandwidth of 0.109
Picking joint bandwidth of 0.123
Picking joint bandwidth of 0.0829
Picking joint bandwidth of 0.0795
Picking joint bandwidth of 0.0533
Picking joint bandwidth of 0.157
Picking joint bandwidth of 0.158
Picking joint bandwidth of 0.0927
Picking joint bandwidth of 0.083
Picking joint bandwidth of 0.101
Picking joint bandwidth of 0.0698

Patient 12 – Pixel Intensity Distributions per Marker

Arcsinh-Transformed, Cofactor = 1



Summary Table: Marker Intensity and Sparsity

```
marker_summary <- pixels_long %>%
  group_by(marker) %>%
  summarise(
    median_intensity = median(intensity_asinh),
    pct_zero         = mean(intensity == 0) * 100, # zero check on raw counts
    max_intensity    = max(intensity_asinh),
    .groups          = "drop"
  ) %>%
  mutate(across(where(is.numeric), ~ round(., 2))) %>%
  arrange(desc(pct_zero))

print(marker_summary)
```

```
## # A tibble: 41 x 4
##   marker      median_intensity pct_zero max_intensity
##   <fct>          <dbl>      <dbl>      <dbl>
## 1 GATA3(Eu151Di)          0      90.8          3.7
## 2 TIGIT(Tm169Di)          0      87.8          4.86
## 3 RORgT(Nd146Di)          0      85.5          2.88
## 4 LAG3(Eu153Di)          0      84.7          3.88
## 5 CCR7(Nd148Di)          0      83.0          7.71
## 6 CD39(Er167Di)          0      76.4          4.36
## 7 CTLA4(Yb176Di)          0      75.4          7.96
## 8 CD117(Dy164Di)          0      74.2          7.88
## 9 CD56(Sm152Di)          0      73.1          3.83
## 10 TBet(Nd145Di)          0      69.3          4.01
## # i 31 more rows
```

```
knitr::kable(marker_summary, caption = "Patient 12 Marker Intensity and Sparsity Summary (asinh-transformed)")
```

Table 1: Patient 12 Marker Intensity and Sparsity Summary (asinh-transformed)

marker	median_intensity	pct_zero	max_intensity
GATA3(Eu151Di)	0.00	90.84	3.70
TIGIT(Tm169Di)	0.00	87.77	4.86
RORgT(Nd146Di)	0.00	85.50	2.88
LAG3(Eu153Di)	0.00	84.67	3.88
CCR7(Nd148Di)	0.00	82.96	7.71
CD39(Er167Di)	0.00	76.42	4.36
CTLA4(Yb176Di)	0.00	75.38	7.96
CD117(Dy164Di)	0.00	74.22	7.88
CD56(Sm152Di)	0.00	73.09	3.83
TBet(Nd145Di)	0.00	69.33	4.01
PD1(Ho165Di)	0.00	67.48	3.64
CD86(Gd155Di)	0.00	64.04	3.99
PDL1(Nd150Di)	0.00	62.37	6.24
CD163(Sm147Di)	0.00	62.17	4.75
CD8a(Gd160Di)	0.00	60.29	4.83
CD19(Nd142Di)	0.00	51.94	6.66
CD11c(Dy161Di)	0.88	42.11	5.39
MKI67(Yb172Di)	0.88	39.13	6.28

marker	median_intensity	pct_zero	max_intensity
CD141(Er170Di)	0.88	38.20	6.15
CD4(Gd156Di)	0.88	37.39	5.69
SMA(Pr141Di)	0.91	34.95	7.74
CD127(Pt195Di)	1.01	29.44	5.86
CD45RA(Er166Di)	1.08	29.33	7.35
LAMP3(Dy163Di)	1.06	28.49	6.29
CD15(Sm149Di)	1.17	26.53	5.74
EpCAM(Nd144Di)	1.56	25.23	5.00
CD138(Er168Di)	1.49	25.20	5.80
TIM3(Nd143Di)	1.67	16.86	7.94
CD45RO(Yb173Di)	2.26	13.06	7.07
CD68(Tb159Di)	1.98	12.50	7.01
TTF1orCD16(Pt194Di)	1.91	11.85	5.43
HLADRB(Yb171Di)	2.44	11.29	8.85
CD28(Yb174Di)	1.91	9.98	6.88
FoxP3(Dy162Di)	2.07	8.50	5.42
CD103(Gd158Di)	2.10	8.38	6.49
DNA191(Ir191Di)	3.53	7.64	7.23
CD38(Pt198Di)	2.07	4.14	5.53
DNA193(Ir193Di)	4.14	3.43	7.94
Membrane(Pt196Di)	3.04	3.19	6.69
CD14(Lu175Di)	2.85	2.02	7.43
CD1c(Sm154Di)	3.14	1.34	6.64

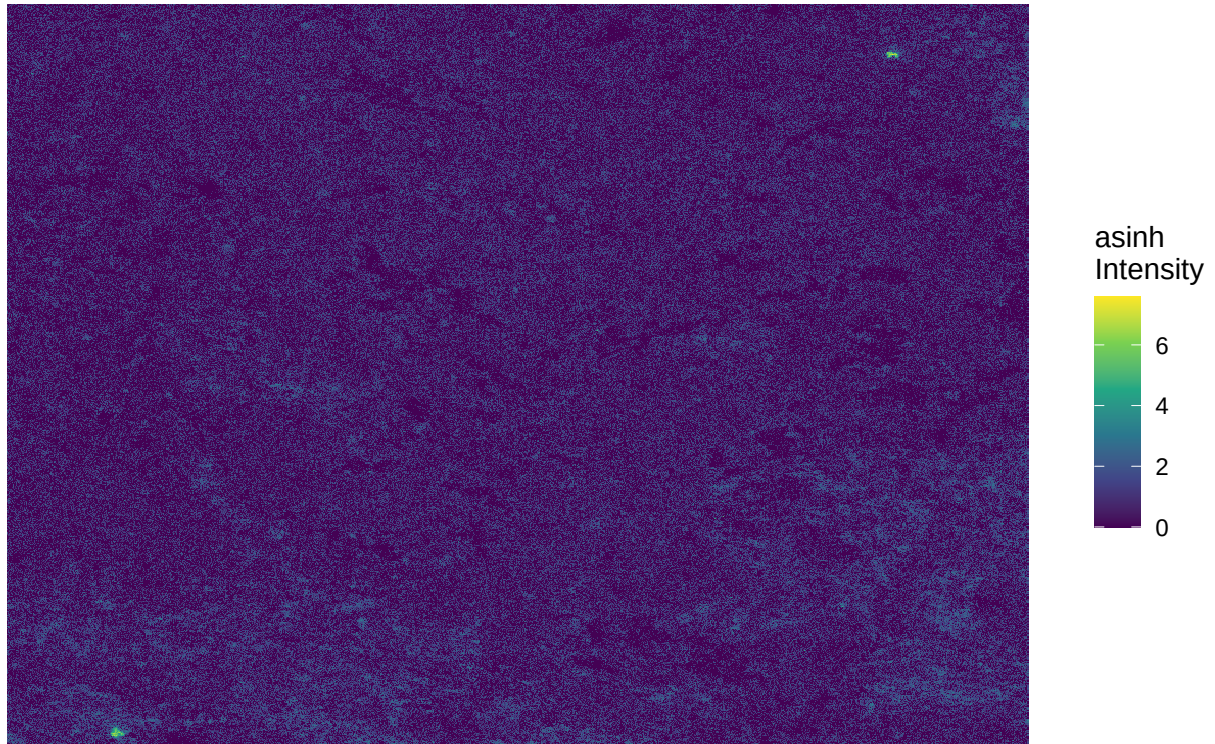
Visualize one channel image (using the pixel coordinates)

```
library(ggplot2)

# Read ROI 1
roi1 <- read.delim(txt_files[1], check.names = FALSE)

# Visualize CD19 as a spatial heatmap
ggplot(roi1, aes(x = X, y = Y, fill = asinh(`CD19(Nd142Di)` / 1))) +
  geom_tile() +
  scale_fill_viridis_c() +
  labs(title = "CD19 - ROI 1", fill = "asinh\nIntensity") +
  theme_void()
```

CD19 – ROI 1



There are two very small bright spots in the lower left and upper right... potential artifacts?

Visualize a selection of immune markers (ROI 1)

```
# Downsample to every 3rd pixel in X and Y (~11% of pixels) for speed
set.seed(42)
roi1_sub <- roi1 %>%
  filter(X %% 3 == 0, Y %% 3 == 0)

cat("Downsampled pixels:", nrow(roi1_sub), "\n")
```

```
## Downsampled pixels: 111556
```

```
# Focused marker list
focused_markers <- c(
  "EpCAM(Nd144Di)",
  "SMA(Pr141Di)",
  "FoxP3(Dy162Di)",
  "CD68(Tb159Di)",
  "CD163(Sm147Di)",
  "CD1c(Sm154Di)",
  "TIM3(Nd143Di)",
  "PD1(Ho165Di)",
  "TTF1orCD16(Pt194Di)"
)
```

```
# Reshape to long format
```

```

roi1_focused <- roi1_sub %>%
  select(X, Y, all_of(focused_markers)) %>%
  melt(id.vars = c("X", "Y"),
       variable.name = "marker",
       value.name = "intensity")

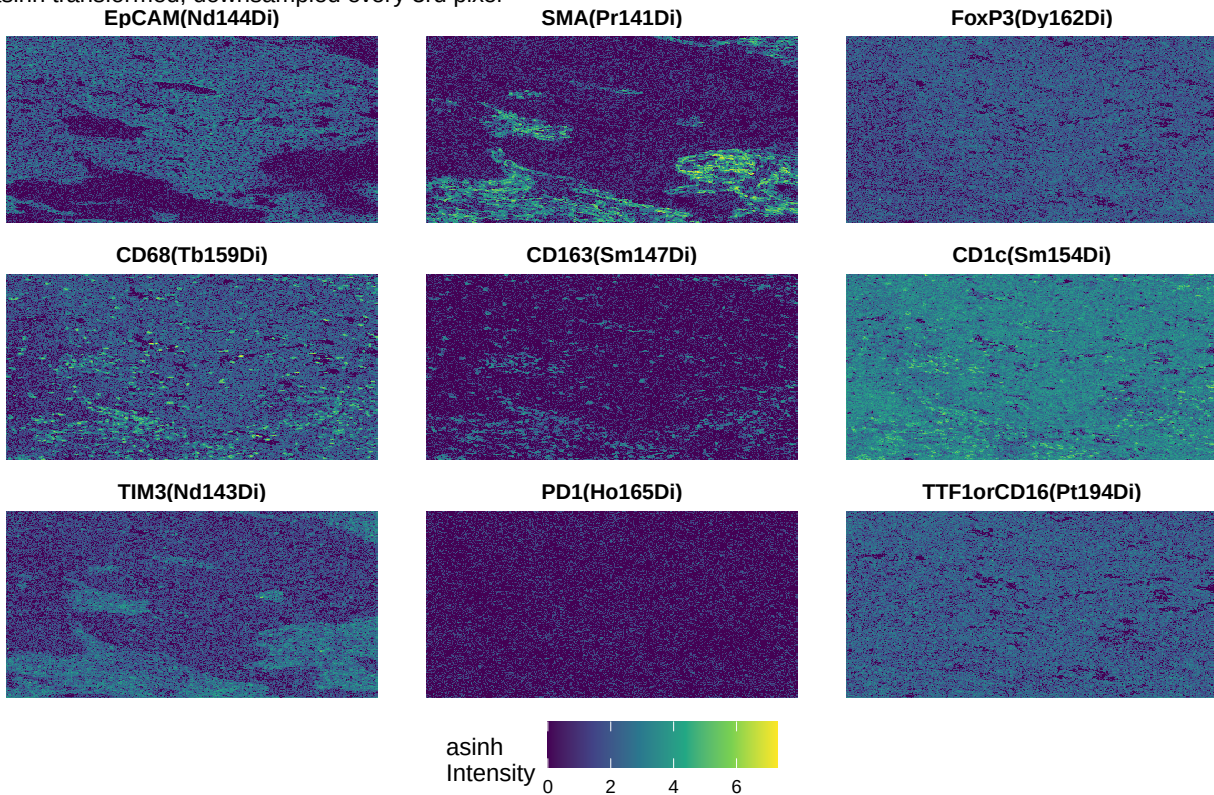
# Plot
p <- ggplot(roi1_focused, aes(x = X, y = Y, fill = asinh(intensity / 1))) +
  geom_tile() +
  scale_fill_viridis_c(
    option = "viridis",
    limits = c(0, quantile(roi1_focused$intensity_asinh, 0.99, na.rm = TRUE)),
    oob = scales::squish # values above the limit get clamped to max color
  ) +
  facet_wrap(~ marker, ncol = 3) +
  labs(
    title = "Patient 12 ROI 1 - Selected Immune Markers",
    subtitle = "asinh transformed, downsampled every 3rd pixel",
    fill = "asinh\nIntensity"
  ) +
  theme_void(base_size = 9) +
  theme(
    strip.text = element_text(size = 8, face = "bold"),
    legend.position = "bottom"
  )

#ggsave("Patient12T_ROI1_focused_markers.pdf", p, width = 12, height = 14)
print(p)

```

Patient 12 ROI 1 – Selected Immune Markers

asinh transformed, downsampled every 3rd pixel



Even after capping the color scale at the 99th percentile (to identify extreme outliers aka hot pixels), the plots still look the same. This probably means that the data doesn't have extreme hot pixels in these 9 markers (unlike CD19). Suggests these channels are relatively clean.

Takeaway: pixel-level visualization is most useful for structural markers like EpCAM and SMA and markers with broad expression. For rare immune populations like FoxP3, the signal is too sparse at the pixel. Steinbock is needed to segment cells first and then sum up the signal per cell, which will make those populations much more detectable.